Knowledge Transfer via Dynamic Policy Fusion between Autonomous Driving Agents

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***Abstract*—Autonomous driving in mixed and dynamic environments requires the ability to learn from diverse traffic scenarios, specifically unsignalized intersections, while adapting to new ones with minimal exploration. In this work, we address the challenge of transferring experience between autonomous driving agents in a Fine-tuned Transfer Learning framework. We consider a setting where a source agent, an expert vehicle previously trained to handle a specific unsignalized intersection, shares its learned policy with a target agent which serves as the ego vehicle and has never encountered the new environment. Rather than relying solely on random exploration, the target agent dynamically integrates the source agent's policy parameters without any additional learning process. We propose a novel policy sharing framework relying on a fusion mechanism for Q-Learning policies between source and target agents. Simulation results in SUMO demonstrate improved efficiency, safer interactions, less collisions and more consistent decision-making compared to related approaches. This work highlights the potential of context-aware policy sharing for accelerating adaptation and enhancing safety in collaborative autonomous driving systems.**

***Index Terms*—Connected and Automated Vehicles (CAV), Autonomous Vehicles (AV), Transfer Learning (TL), Reinforcement Learning (RL), Dynamic policy transfer.**

I. Introduction

Autonomous Vehicles (AVs) are expected to reshape transportation by addressing the critical issue of human error, which contributes to 94% of crashes (NHTSA [1]). While AV technologies promise improved safety, lower emissions, and better mobility [1], their deployment in complex urban environments requires robust real-time systems combining perception, localization, planning, and control with advanced sensors and computing [2]. Also commonsense knowledge can help autonomous vehicles make smarter, more human-like decisions, improving smart city mobility and encouraging teamwork between AI and Law [3]. Recent regulatory developments highlight the need for standardized evaluation frameworks [4]. In this context, effective knowledge-sharing between AVs appears essential for adaptive decision-making in dynamic traffic scenarios [2], enabling collaborative learning while maintaining operational safety. Multi-agent Reinforcement Learning (MARL) allows autonomous vehicles to learn collaboration in dynamic settings by modeling their interactions and using decentralized optimization to balance individual and group goals like traffic flow and safety [5], with key challenges including agent selection and role heterogeneity, as seen in ramp-merging scenarios where merging vehicles learn actively and highway vehicles observe passively [6].

Beyond multi-agent settings, knowledge transfer across policies, particularly in cross-domain scenarios, is increasingly studied [7], as reusing knowledge from related tasks can significantly accelerate learning and improve generalization. Using pre-trained policies from other tasks or vehicles allows AVs to adapt more efficiently to unseen environments while reducing data requirements [8]. Transfer reinforcement learning approaches such as fine-tuning [8], meta-RL [9], and policy distillation [10] have shown promise, yet they face persistent challenges including catastrophic forgetting, domain mismatch, and expert selection. An open research question remains dynamic policy sharing: deciding what, when, and with whom to transfer knowledge in order to enhance both learning efficiency and operational safety [11].

This work presents a novel cooperative learning framework for autonomous vehicles that enables efficient dynamic adaptation to new environments between autonomous agents via Q-value fusion. Our method facilitates dynamic policy sharing across distinct driving tasks, more specifically from intersection to ramp merge and vice versa. Our experiments in a microscopic traffic simulator demonstrate enhanced safety, with a significant 26% reduction in collision rates, alongside improved efficiency marked by faster convergence to optimal speeds with minimal collisions, compared to no-transfer baselines.

II. Related work

Reinforcement learning (RL) and deep reinforcement learning (DRL) have become essential for autonomous vehicles, enabling them to learn complex driving behaviors and high-level planning strategies [12]. By learning directly from interactions with the environment, RL-based approaches allow AVs to make adaptive decisions in dynamic and uncertain traffic scenarios, improving robustness and efficiency. In particular, transfer learning within RL has attracted increasing attention as a means to accelerate policy adaptation across different tasks, environments, or vehicle platforms, while improving generalization and sample efficiency. Several recent studies illustrate diverse approaches to transfer RL in autonomous driving. For instance, Hou et al. [13] proposed an evolutionary transfer reinforcement learning framework inspired by Darwinian evolution and memetics, enabling multi-agent systems to share and adapt knowledge more effectively, outperforming existing transfer learning methods in complex environments. Yu et al. [14] proposed an intelligent vehicle trajectory planning method based on deep RL and model transfer, which directly generates optimal continuous control action sequences by training a DDPG model in an abstract vehicle dynamic environment and transferring knowledge to real-world scenes. Beyond driving, Lei Cai et al. [15] introduced a multi-agent transfer-RL approach for collaborative target recognition in underwater vehicles, fusing features and affine invariance, and transferring learned feature representations across domains to reduce computation while maintaining high accuracy.

For instance, traffic congestion reduction was addressed through a privacy-preserving asynchronous federated learning algorithm for vertically partitioned data, improving traffic flow, signal waiting time, and fuel efficiency [16]. Travel time optimization was tackled with a cooperative multi-agent FRL algorithm that enhances routing efficiency and reduces delays through distributed and collaborative learning [17]. Collision avoidance is a primary goal in FRL for CAVs. Collaborative learning strategies help agents anticipate and prevent collisions in dynamic environments. For instance, a distributed coordinated brake control algorithm was proposed to enable multiple connected automated vehicles to cooperatively avoid longitudinal collisions [18]. Our approach also targets collision avoidance but emphasizes a safety-efficiency trade-off, minimizing travel time while maintaining context-aware speeds.

These objectives have been studied in various complex driving scenarios, with lane changing and signalized intersections receiving particular focus due to their frequency and complexity in urban and highway settings. Luo et al. [19] developed a lane-change model using Vehicle-to-Vehicle (V2V) data-exchange to decide whether to proceed or abort a maneuver based on a minimum safety spacing (MSS) criterion. However, their model is limited to two-lane scenarios and does not support multi-vehicle coordination. Similarly, Xu et al. [20], formulate cooperative merging as an optimization problem to minimize travel time and maximize merging vehicles. Using a genetic algorithm, the approach improves traffic efficiency and reduces fuel consumption, but lacks V2V knowledge transfer, limiting adaptability to dynamic scenarios. Signalized intersections are critical for traffic flow and safety, yet most current traffic signal systems use fixed or simple adaptive controls that lack real-time intelligence. To overcome these limitations, recent approaches leverage CAVs with reinforcement learning. By sharing traffic data and employing learning-based car-following models, CAVs can adapt their driving behavior dynamically at intersections, enhancing travel efficiency, lowering energy use, and improving safety in real time [21].

Unlike signalized intersections with established control, unsignalized intersections require vehicle coordination and situational awareness. Reinforcement learning (RL) offers a promising solution, enabling vehicles to learn safe, efficient, and cooperative behaviors in dynamic traffic [22]. Shu et al. [23] proposed a deep transfer RL framework that reuses decision strategies across intersection tasks, improving efficiency, safety, and real-time performance.

Cooperative learning via policy sharing accelerates adaptation and enhances robustness, while policy fusion combines multiple strategies for stronger decision-making, often guided by RL-based policy ranking [24]. Mix Q-Learning for Lane Changing (MQLC) further coordinates global and individual Q-Networks with intent recognition to balance individual and collective benefits, though its scope is limited to lane-changing and excludes tasks like collision avoidance and environment adaptation [25]. In this work, we address these challenges by proposing a dynamic reinforcement learning framework based on knowledge transfer via policy fusion aimed at enabling intelligent and adaptive decision-making in two types of intersections. T-intersections involve negotiation and right-of-way decisions, while ramp merges require smooth and safe integration into ongoing traffic both demanding context-aware and flexible strategies. Our approach allows CAVs to generalize learned policies to unseen intersection scenarios without retraining. This enhances the model by dynamically integrating source agents' policy parameters based on traffic context similarity, enabling faster convergence, safer interactions, and more consistent decisions compared to the baseline.

III. Methodology

Our study investigates Peer-to-peer knowledge sharing between autonomous vehicles in cross-environment transfer scenarios. We examine two agents: the Red vehicle (expert in T-intersections, as illustrated in Fig. 3(a)) and the Orange vehicle (expert in ramp merging, Fig. 3(b)). The experiments focus on two transfer learning protocols:

(i) Basic transfer learning, agents are introduced to new environments and initially rely on their previously learned policy for navigation in a different environment. They undergo 100 more training episodes in the new setting without any guidance from an expert agent, which is assumed to be insufficient for effective adaptation to the new environment.

(ii) Transfer learning combined with fine-tuning using source agent guidance. This protocol represents our major contribution. In this approach, we introduce a novel policy fusion mechanism that enables direct knowledge sharing between the source and target agents without requiring any additional training. We compare these protocols with four baselines:

(1) The Q-Learning (QL) expert agent refers to an agent that has undergone extensive training and evaluation in the target environment, with 1000 episodes, allowing it to develop and refine an effective policy for ideal performance.

(2) The Deep Q-Network (DQN) expert agent that undergone 1000 episodes training and evaluation in the target environment.

(3) Zero-shot QL (ZS-QL) involves using the pre-trained QL model from the original environment and directly testing it with a 100 episodes in the new environment without any additional training.

(4) Zero-shot DQN (ZS-DQN) includes the testing of the pre-trained DQN with 100 episodes in the new environment.

In this context, an episode is defined as the interval beginning when the vehicle enters the simulation and ending either when it successfully reaches its destination and exits the simulation, or when it exceeds a predefined time limit for completing the task.

A. Problem formulation

We model the considered intersection problem as a Markov decision process (MDP) as illustrated in Fig. (1). We introduce the proposed Q-Learning algorithm that solves the formulated MDP effectively.

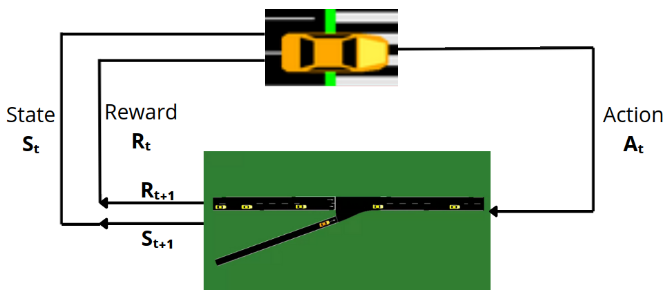


Fig. 1: Our MDP problem

MDPs provide a mathematical framework for modeling decision-making in environments where outcomes are partially stochastic and partially controllable by an agent. An MDP is formally defined by the tuple $(\mathcal{S}, \mathcal{A}, P, R, \gamma)$, where:

- $\mathcal{S}$ represents the set of all possible states in the environment.
- $\mathcal{A}$ denotes the set of all possible actions available to the agent.
- $P(s'|s,a)$ is the transition probability function that specifies the likelihood of moving to a particular next state s' from state s after taking action a.
- $R(s,a,s')$ is the reward function, which specifies the immediate reward received after transitioning from state s to state s' by taking action a.
- $\gamma \in [0,1)$ is the discount factor, which determines the present value of future rewards.

The goal of an MDP is to find an optimal policy $\pi^* : \mathcal{S} \to \mathcal{A}$ that optimizes the total expected reward accumulated over time. The action-value function $Q^\pi(s,a)$ denotes the expected total reward obtained by starting in s, taking action a, following the given policy π:

$$\mathrm{Q}^\pi(s,a) = \mathbb{E}\left[\sum_{t=0}^{\infty} \gamma^t R(s_t, a_t, s_{t+1}) \middle| s_0 = s, a_0 = a, a_t \sim \pi(s_t)\right]$$

B. Q-Learning

QL is a model-free RL algorithm that aims to learn the optimal action-value function $Q^*(s,a)$, which represents the maximum expected cumulative reward achievable from state s by taking action a and then following the optimal policy π^*. The optimal policy π^* is derived from the optimal action-value function as follows:

$$\pi^*(s) = \arg\max_{a \in \mathcal{A}} Q^*(s,a). \tag{1}$$

QL iteratively updates the action-value function using the Bellman equation. The update rule is given by:

$$Q(s,a) \leftarrow Q(s,a) + \alpha\left[R(s,a,s') + \gamma \max_{a'} Q(s',a') - Q(s,a)\right] \tag{2}$$

where:

- $\alpha \in (0,1]$ is the learning rate, which controls the weight given to new information.
- $R(s,a,s')$ is the immediate reward received after transitioning from state s to state s' by taking action a.
- γ is the discount factor, which determines the importance of future rewards.
- $\max_{a'} Q(s',a')$ is the maximum estimated future reward from the next state s'.

The QL algorithm is off-policy, meaning it learns the optimal policy π^* while following an exploratory behavior policy. We employ ϵ-greedy exploration.

$$a_t = \begin{cases} \text{Random action} & \text{if } x > \epsilon \\ \text{Best action } (Eq.(1)) & \text{otherwise} \end{cases} \tag{3}$$

where $x \sim \mathcal{U}(0,1)$ a random number generated at each decision step. This property allows QL to explore the environment effectively while still converging to the optimal action-value function Q^*.

C. Reinforcement Learning Setup

This section describes our RL setup describing the State Space, Action Space, State Transition and Reward Function.

State Space: In our simulation, the environment returns a state vector at each time step t, representing various features of the vehicle. The state is defined as:

$$S_t = (gap, \textit{safe distance}, V_t),$$

where *gap* represents the Euclidean distance between the position of the ego vehicle and the leading vehicle. *Safe distance* is the minimum distance required to avoid a collision if the leading vehicle brakes suddenly, following the 2-second rule. Specifically, the safe distance is calculated as safe distance $= 2 \times V_t$, where V_t is the current velocity of the ego vehicle. Maintaining an appropriate velocity is crucial for safety, as higher speeds not only increase the required stopping distance but also amplify the severity of potential collisions (since kinetic energy scales with V_t^2). The acceleration A_t is also an important parameter because of its impact on the dynamic: positive acceleration reduces the gap more quickly, while negative acceleration (braking) must be sufficient to maintain or restore a safe distance if the leading vehicle decelerates. Therefore, both V_t and A_t must be carefully controlled to ensure that the gap always remains above the safe distance, minimizing the risk of collisions.

We discretize our environment based on the overall state perceived by the vehicle. For instance, a black state represents a collision, red indicates highly dangerous navigation, orange corresponds to moderately dangerous navigation, green signifies safe navigation, and white means the goal has been reached without any collision. The level of danger is determined according to how well the vehicle maintains the safe distance; states are classified based on whether the current gap respects or violates the required safety threshold. This discretization approach is environment-independent and has proven to be the most effective among all the discretization techniques we tested.

Action Space: The agent controls the vehicle using discrete acceleration commands. At each step, it can choose to *accelerate*, *maintain speed*, or *decelerate*.

State Transition: We model the environment as a MDP, where the probability of transitioning to a new state s_{t+1} given the current state s_t and action a_t is expressed as:

$$P(s_{t+1} \mid s_t, a_t)$$

Reward Function: The agent receives a scalar reward signal after each action to guide its learning process. The reward function (4) is defined as:

$$\text{R} = \begin{cases} -20 & \text{collision occurs} \\ -5 & \text{dangerous navigation} \\ -1 & \text{if vehicles are too close} \\ 5 & \text{safe navigation} \\ 50 & \text{destination successfully reached} \\ & -0.01 \times \text{waiting time} \end{cases} \tag{4}$$

The vehicle should reach its destination as efficiently as possible, minimizing travel time, maintaining a safe following distance and avoiding collisions with other vehicles along its route.

D. Deep Q-Network (DQN) Baseline

We use a Deep Q-Network as a baseline. Unlike tabular QL, which cannot scale to continuous state spaces, DQN uses a neural network to approximate the action-value function $Q(s, a)$. This allows the agent to learn directly from the continuous state vector.

State space : is now a vector of continuous states.

$$S_t = (V_t, gap, \textit{waiting time}),$$

The DQN agent samples transitions from a replay buffer and updates its network by minimizing the temporal-difference loss.

$$\text{L}(\theta) = \mathbb{E}_{(s,a,r,s')\sim D}\left[\left(r + \gamma \max_{a'} Q(s', a'; \theta^-) - Q(s, a; \theta)\right)^2\right]$$

where:

- D is the experience replay buffer, which stores past transitions (s, a, r, s') to break the correlation between consecutive samples,
- θ^- are the parameters of a target network, periodically updated to stabilize training,
- γ is the discount factor.

The action space remains discrete: *accelerate*, *maintain speed*, or *decelerate*.

The reward function is adapted for continuous states and penalizes collisions, risky proximity, and excessive waiting time, while rewarding efficient and safe goal achievement. It remains the same as described in equation (4).

E. Proposed approach: dynamic QL policy fusion

To enable policy generalization by eliminating the inefficiency of random exploration during learning in a new environment, we propose a dynamic transfer strategy based on Q-table fusion using a confidence coefficient ε_t from ε_{greedy} in equation (3). In our framework, instead of relying on stochastic actions, the target agent incorporates prior knowledge from a source agent that has already mastered a related driving task. This allows the agent to act meaningfully from the start, even in a previously unseen environment.

Let $Q_s(s, a)$ and $Q_t(s, a)$ represent the Q-values of the source and target agents for a given state-action pair (s, a). The fused Q-value used for decision-making is given by:

$$Q_{\text{fused}}(s, a) = (1 - \varepsilon_t) \cdot Q_t(s, a) + \varepsilon_t \cdot Q_s(s, a) \tag{5}$$

where $\varepsilon_t \in [0, 1]$ is a time-dependent coefficient representing the growing confidence in the target agent's policy. Initially, $\varepsilon_0 \approx 1$, which prioritizes the source model. As the target agent gains experience in the new environment, ε_t decreases accordingly (Fig. 2).

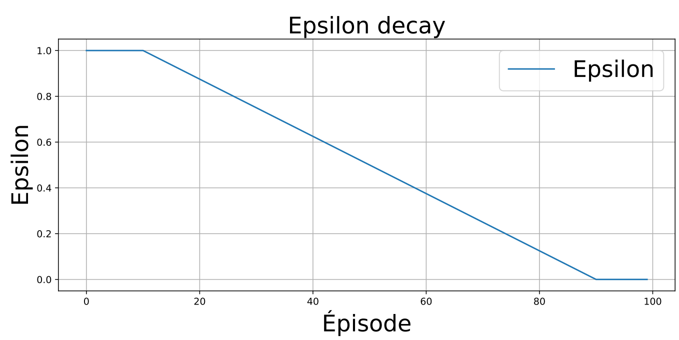


Fig. 2: Epsilon decay

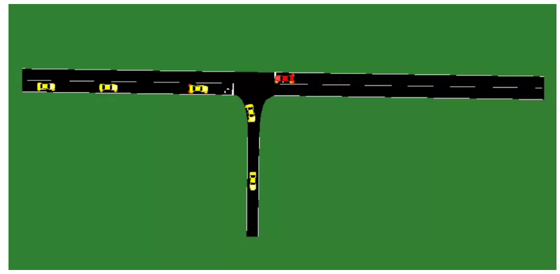
The results show that this method enables a smooth and efficient transfer of knowledge. It avoids unsafe or inefficient random exploration, and instead provides the target agent with meaningful guidance without needing any more training. Over time, the agent gradually builds a reliable policy adapted to its new environment, while using previously gained expertise from the expert.

IV. Experimental Setup

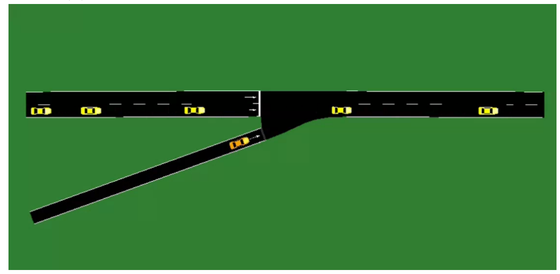
This section presents the experimental setup used to evaluate our RL agents. We utilize a micro-traffic simulator to create and control diverse traffic scenarios. The experiments focus on two critical situations for autonomous vehicles: T-intersections and ramp merges, both of which pose significant challenges for decision-making and safety. For each experiment, we report the average results over three independent runs. This approach ensures that our findings are robust and not influenced by random fluctuations or outlier events in a single trial.

A. SUMO Simulator

We conduct our experiments using Simulation of Urban Mobility (SUMO) [26] to model our environment. SUMO comes with a Traffic Control Interface (TraCI), a TCP-based API that facilitates real-time interaction between external programs and the simulation environment. SUMO has been widely used in the literature as a foundation to advanced traffic control frameworks. One notable example is FLOW [27], an open-source toolkit that integrates SUMO with deep reinforcement learning libraries like RLlib. FLOW enables the design and training of autonomous vehicle and infrastructure controllers in diverse traffic scenarios, demonstrating how SUMO can support advanced AI-driven traffic research. In our study, this interface enables tight integration between the AV agents and the simulated world, allowing each agent to receive environment feedback and issue driving actions at each time step.



(a) Intersection scenario with the red vehicle



(b) Merging scenario with the orange vehicle

Fig. 3: Illustration of our two reinforcement learning tasks

B. Experimental Scenarios: T-Intersections and Ramp Merges

Intersections can be categorized into several main types based on their geometric topology, such as crossroads, X-intersections, Y-intersections, T-intersections, roundabouts, misaligned intersections, ramp merges, and deformed intersections [28]. In our study, two autonomous agents navigate distinct driving contexts: T-intersections and Ramp merge . In both scenarios, agents operate within dense and dynamic traffic flows generated using SUMO vehicle generation system with randomized time-gaps. This setup ensures diverse and unpredictable conditions that closely mimic real-world urban mobility. Each agent learns to navigate its respective scenario independently while adapting to surrounding varying traffic densities. To illustrate the learning tasks tackled by each agent, we present two representative scenarios extracted from our simulation environment. Figure 3(a) represents the intersection task, where a red vehicle learns to navigate a T-shaped urban intersection from scratch, without prior knowledge or external guidance. The agent must learn to adjust appropriately, handle crossing traffic, and complete its turn safely under variable traffic densities. Figure 3(b) presents the merging task, in which an orange vehicle independently learns how to merge correctly. This scenario requires the agent to develop strategies for acceleration, gap selection, and safe insertion into a high-speed traffic stream. In both cases, the learning process is driven by RL, where the agents must explore, interact with surrounding vehicles, and refine their policies over time to achieve successful navigation. These two tasks serve as the basis for evaluating our dynamic policy-fusion approach.

After training, both Red and Orange agents have learned from scratch their policy to navigate on their respective environments with 1000 episodes using QL and DQN. They learned to adapt their behavior to dynamic traffic conditions, exhibiting improved decision-making by maintaining appropriate speeds and safe inter-vehicle distances throughout their respective navigation tasks. As shown in Tab. I and Fig. 4, the overall reward metrics are very similar for both tabular QL and DQN. However, in terms of episode success rate, the tabular QL agent outperforms DQN, demonstrating better efficiency in reaching the goal without collisions under the same training conditions.

TABLE I: Episode Success of Red and Orange experts

	QL	DQN
Red agent in Intersection	91.67	87
Orange agent in Merge	97.67	88

Fig. 4: Mean Reward of Red and Orange experts

V. Results

In this section, we explore our multi-agent cooperative learning scenario in which the two autonomous agents are exposed to novel driving environments without prior direct experience. Specifically, the red agent, originally trained in T-intersection navigation, is introduced to a ramp merging scenario (Fig. 3(b)), while the orange agent, initially trained on merging tasks, is placed in a T-intersection context (Fig. 3(a)). The core challenge lies in enabling each vehicle to adapt effectively to the new environment without retraining from scratch, nor needing a few more examples to learn. To address this, we propose a dynamic policy fusion approach that leverages cross-agent experience. Instead of relying solely on random exploration, the agent utilizes learned policies from the expert agent to guide its decision-making in the new task. This is achieved through the model fusion mechanism applied to the QL networks of the target agent (eq. (5)). By integrating the learned Q-values, the target agent is able to benefit from the source agent's prior experiences, thus accelerating learning and reducing unsafe or suboptimal actions during exploration. Our method enables generalization by transferring task-relevant knowledge across distinct but structurally related driving scenarios. To evaluate generalization, we report validation curves for key performance metrics such as reward, average speed, three following vehicles deceleration and episode success rate (Fig. 5 and 6; Tab. II and III). These experiments aims to highlight the benefit of cross-agent knowledge transfer via policy fusion versus minimal direct exposure. Our goal to achieve a good compromise between efficiency and safety. The episode success rate metric provides a direct assessment of safety, since an episode is considered a failure if a collision occurs. In contrast, the other traffic-related metrics presented offer insight into traffic efficiency.

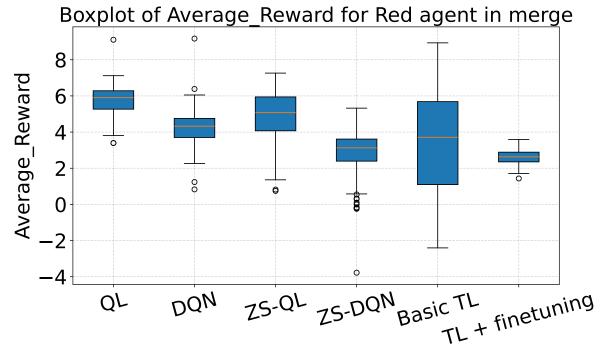
A. Red agent in Ramp merge

Our first experiment involves evaluating the red agent, which was initially trained on T-intersection navigation, in a novel environment: a ramp merging scenario. We consider two approaches for adaptation.

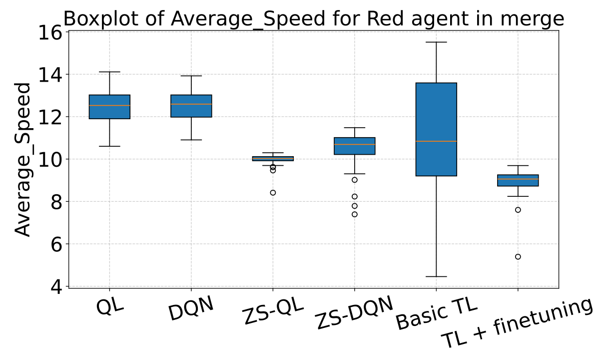
The first is Basic Transfer Learning (TL): Red Agent is introduced to new environment (Ramp merge) and initially rely on its pre-trained policy for navigation. Only 100 training episodes in the new environment were added, without any guidance from an expert agent. The second is Transfer Learning with Fine-Tuning (TL + fine-tuning). In this protocol, target agent (Red) benefits from source agent guidance (Orange) via our model fusion mechanism (Section III-E). It is utilized during fine-tuning to facilitate policy fusion and improve adaptation to the new environment.

We compare this to the performance of the Expert Agent (QL and DQN) and Zero-Shot Learning (ZS-QL and ZS-DQN) where The pre-trained model from the original environment is directly tested in the new environment without any additional training. This comparison highlights the effectiveness of cross-agent knowledge transfer versus minimal exposure-based learning.

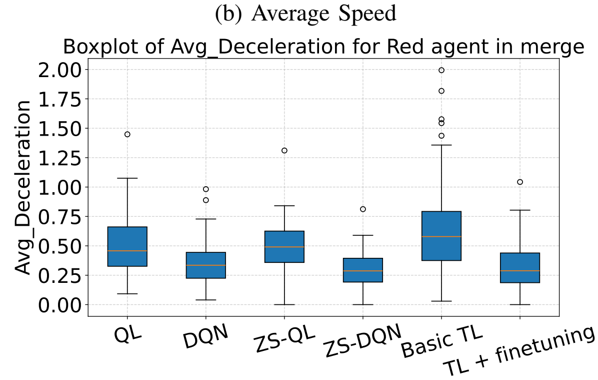
The results indicate that transitioning from intersection to merging is challenging without assistance and requires additional training, whereas knowledge transfer via policy sharing from the source agent enables faster and easier adaptation.



(a) Reward



(b) Average Speed



(c) Average Deceleration

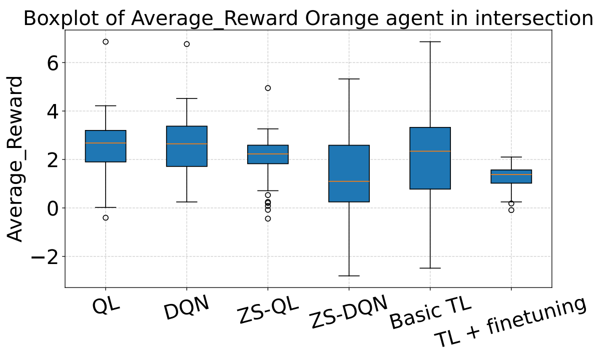
Fig. 5: Validation metrics of the Red agent in the ramp merge scenario

B. Orange agent in Intersection

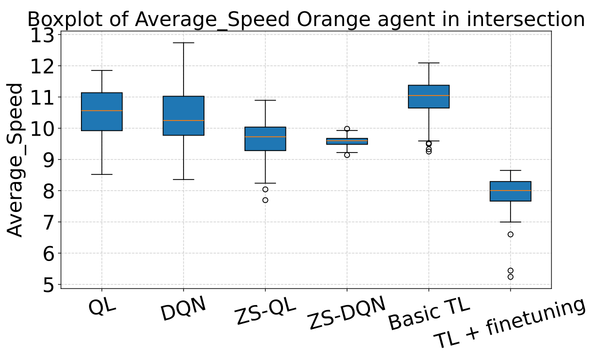
In the second experiment, we reverse the roles: the orange agent, originally trained on ramp merging, is now evaluated in a new environment: T-intersection.

Similarly as the Red agent, we analyse Orange agent metrics by comparing four adaptation strategies with or without external help.

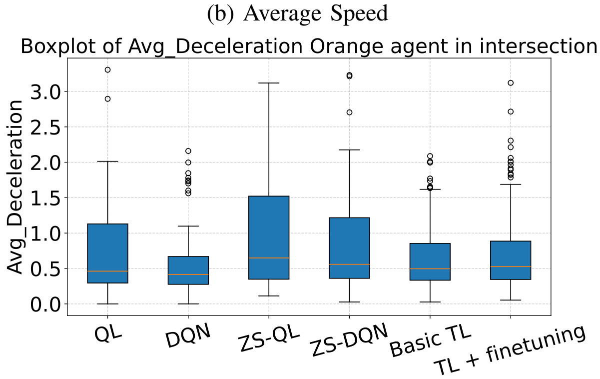
The results show that adapting from a merging scenario to an intersection task is more challenging, as the intersection environment is inherently more complex. This increased difficulty makes it harder for the vehicle to adapt when transferring knowledge from merging to intersection, highlighting the value of cooperative learning in facilitating this transition.



(a) Reward



(b) Average Speed



(c) Average Deceleration

Fig. 6: Validation metrics of the Orange agent in the intersection scenario.

C. Discussion

Our success rate metric is defined as episodes completed without any collision as shown in Tab. II and III. Focusing on success rate, we observe that zero-shot learning leads to a significant drop in agent performance when the environment changes. Adding 100 additional training episodes in the new environment (basic transfer learning) did not yield any improvement at the intersection, nor the merging scenario, compared to Zero-shot learning. In contrast, our method achieved improvements in both environments: a 26% increase at the intersection and a 14% increase in the merging scenario. We also notice that transitioning from intersection to merging proves easier adaptation than the opposite.

TABLE II: Success rates (%) of agents in both scenarios without cooperation using Zero-shot and basic TL.

(a) Zero-shot Learning

	Red Agent	Orange Agent
Scenario 1 (Intersection)	91.67	69.33
Scenario 2 (Merge)	89.34	97.67

(b) Basic Transfer Learning

	Red Agent	Orange Agent
Scenario 1 (Intersection)	91.67	64.33
Scenario 2 (Merge)	85.67	97.67

TABLE III: Success rates (%) of agents in both scenarios with cooperation using TL and policy fusion fine-tuning.

	Red Agent	Orange Agent
Scenario 1 (Intersection)	91.67	91
Scenario 2 (Merge)	100	97.67

Focusing on the Red agent in the ramp merge scenario (Fig. 5), we observe that the reward curve for our method (Fig. 5(a)) is lower overall, which is explained by the agent maintaining a lower average speed. However, this reward is also the most stable and exhibits the least variability compared to both the scenario without assistance and the baseline. Regarding average speed (Fig. 5(b)), the agent without assistance tends to drive at higher speeds, which frequently leads to collisions—several of which are observed in the absence of help, while no collisions occur when assistance is provided. This difference is largely due to the higher speeds adopted by vehicles operating without help. Additionally, traffic flow is smoother when in our experiment compared to Basic TL: fewer vehicles are forced to decelerate abruptly (more than 0.5 m/s^2), whereas without help, up to three vehicles occasionally experience such strong braking (Fig. 5(c)).

A similar pattern is noted for the Orange agent (Fig. 6), the reward with Red agent assistance is lower but more stable (Fig. 6(a)). Figure 6(b) shows that our model quickly adapts to the new environment by maintaining stable, moderate speeds with less fluctuation, resulting in fewer collisions compared to other methods. The braking of the affected vehicles is comparable but less harsh than other presented methods (Fig. 6(c)).

These experimental results demonstrate that our dynamic policy fusion approach significantly accelerates the adaptation process, enables a gradual transition from the source policy to the target policy and leverages trusted knowledge from the source agent to make safer decisions in unfamiliar scenarios. Compared to agents encountering unfamiliar environments without prior knowledge, agents leveraging the fused Q-table achieve a better balance between efficiency, safety, and rapidity, due to transfer from an expert agent. This not only indicates faster convergence but also improved overall decision-making. Furthermore, the number of successful episodes, as shown in Table II and III is noticeably higher. These findings suggest that the transferred knowledge enhances both safety and efficiency, enabling agents to navigate complex scenarios more reliably. While the current work demonstrates successful policy fusion between homogeneous agents, there remain promising avenues for further development. We plan to extend our policy fusion approach to heterogeneous agents to broaden its applicability, and to develop autonomous task change detection so agents can adapt to new tasks without manual intervention, improving cooperation and scalability in complex environments.

VI. Conclusion

This work presents a dynamic policy sharing framework for multi-agent reinforcement learning in autonomous driving scenarios. By enabling adaptation through QL policies fusion without training, our approach facilitates cross-agent generalization between structurally different environments. Experimental results demonstrate that agents benefit significantly from prior experience shared by their expert peers, improving both adaptation speed and safety. Compared to limited retraining, the proposed method provides more stable performance and reduces critical events such as collisions and sudden braking. These findings highlight the potential of cooperative learning strategies in enhancing the robustness and efficiency of autonomous systems in unfamiliar environments. Future directions include exploring how to effectively select or weight multiple source agents when more than one is available, as well as investigating model fusion techniques when agents do not share the same architecture. This opens promising avenues for domain adaptation and more flexible cross-agent collaboration in real-world deployments.

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